Detailed description of the design including the CONTROL treatment and the POLICY treatment. Please particularly outline the intervention in the POLICY treatment in detail.

After a short intro screen informs about the content of the survey and invites respondents to read the following information carefully, **all respondents** are informed that the consequences of climate change create "social costs" providing some examples for such costs, and stressing that these costs are born by everyone, irrespective of their pollution behavior. All respondents are informed about current estimates of the amount of social costs caused by one additional ton of greenhouse gas emissions in the US (USD 120.00). Emission trading systems are introduced as one possible response to greenhouse-gas emissions. (In the following all text given to respondents (question wording or information) will be highlighted in blue.)

In the following, you will read a brief explanation of a policy that aims to reduce greenhouse gas emissions in the U.S. Please read this information carefully, as you will be asked afterward whether you have understood how the policy works and whether you support it.

The consequences of **climate change create substantial social costs** due to extreme weather events, reduced productivity, and health issues caused by higher temperatures. **These costs are borne by everyone**, regardless of whether an individual produces more or fewer emissions.

Greenhouse gases, such as carbon dioxide, are a primary driver of climate change. The social costs created by one additional ton of greenhouse gas emissions can be expressed in dollar terms: According to the U.S. Environmental Protection Agency, **each ton of emissions generates at least USD 120.00 in social costs.**

While some believe that the government should not intervene by regulating greenhouse gas emissions or that technological advances will ultimately solve these problems, others argue that new policies are necessary to reduce emissions. One such policy proposes introducing an **Emission Trading System**.

On the second screen **all respondents** receive a brief explanation of how Emission Trading Systems work.

How do Emission Trading Systems work?

Emission Trading Systems imply that the **government sets a cap on the total amount of greenhouse gases that can be emitted**. The total number of emission allowances is then divided into permits and sold at a fixed price.
Companies are only allowed to emit as much greenhouse gas as they have permits for. If a company doesn't use all its permits, it can sell the excess. If it needs more, it can buy them from others who have spare permits.
Because **emissions now carry a financial cost**, which is likely to increase the prices of goods produced, companies are incentivized

to develop greener products and processes to remain competitive.
As a result, overall **greenhouse gas emissions are reduced, and the social costs are paid by those who cause them**.

This screen is followed by a screen with two questions, which aim to test attention and understanding of the first screens. All **respondents answer** these attention-check / understanding questions.

Please answer the following two questions on the information you just received.

Under an Emission Trading System, companies must purchase emission permits for the greenhouse gases they produce.
[correct] [not correct]

Under an Emissions Trading System, companies can sell emission permits they did not use.

[correct] [not correct]

The **policy treatment group** receives additional information that shows how the problem of overfishing has been successfully regulated by catch shares – which are essentially equivalent to cap-and trade. The control group does not receive this additional information. The treatment involves 3 screens of text which have to be read by participants. The first screen introduces the problem of overfishing and explains the policy of catch-shares. The second screen informs about the results of a study that evaluated the success of catch-share programs in 15 major fisheries in the U.S. and Canada. Treated respondents are informed about catch-share's impact on the development of the fish-population, discard rates, revenues for fisherman and the total allowed catch. The third screen informs that catch-shares were successful because sustainable practice was aligned with fishermen's economic interest and draws a parallel to Emission Trading Systems.

How fishing permits solved the problem of overfishing:
The strategy of limiting businesses' ability to make individual profits at the expense of everyone else has been successfully used to address the problem of overfishing.
Prior to 1976, **U.S. fisheries were largely unregulated**. Fish in the sea were considered a common-pool resource and fishermen competed to catch as many fish as possible to maximize their profits. However, this practice eventually **led to declining fish populations** in many fisheries, **threatening both the ecosystem and the livelihoods of fishermen**.
In response, the **government began regulating fisheries**:
Fishery **managers set catch limits** for specific fish species. The total amount of fish allowed to be caught was **divided into fishing permits**, which were allocated to individual fishermen. Fishermen could either use their fishing permits or sell them to others.

A comprehensive, independent scientific study of 15 major fisheries in the United States and Canada found that the introduction of **these regulations had several positive effects** on both the ecosystem and the fishing industry:

1. Catch limits have led to **healthier fish populations**, with some stocks recovering to more than 60% above historic lows.
2. As fish populations recovered, the **total amount of fish allowed to be caught increased** by an average of 19% ten years after the policy was implemented.
3. By reducing the race to catch fish and raising the cost of spoilage, permits led to a 66% **reduction in discard rates** over ten years.
4. **Revenues** per vessel **increased significantly.** The higher quality of the fish and sustainable fishing certification allowed fishermen to sell at premium prices.

After fishing companies and the market were unable to regulate themselves, leading to a substantial decline in fish populations, regulations helped restore both the ecological and economic sustainability of U.S. fisheries.
These regulations worked because they made overfishing costly, ultimately aligning the interests of fishermen with sustainable practices.

The **mechanism behind Emission Trading Systems is similar**: those who pollute more face higher costs. In this way, a price on carbon encourages companies to develop greener products and processes, while the revenue from selling emission permits can help cover social costs.

All respondents are then presented with the following a policy proposal for an Emission Trading System:

The policy:
The government sets the total amount of allowed greenhouse gas emissions so that the U.S. can meet its 2030 domestic emissions reduction target. This goal aligns with the objective of limiting global warming to below 2°C. Permits are sold to companies annually and can be traded. The price of emitting one metric ton of greenhouse gasses is set at USD 120.00 (the Carbon Price).

Use of revenues:
Money from selling the permits goes into a relief fund for extreme weather events and supports programs that invest in energy efficiency, clean transportation, solar energy, and other ways to reduce greenhouse gases.

Who has to pay?
The policy will apply to companies in electricity generation, fuel supply, and large industrial sectors. This is likely to result in higher prices for consumer goods.

An average American, who currently emits about 14 tons of CO_2 per year, would have to pay roughly USD 1,700.00 annually, assuming individuals don't change their behavior, and businesses don't innovate to offer less emission-intensive products. The actual impact of the policy on each individual will depend on the individual's consumption habits and the associated greenhouse gas emissions.

After the measurement of the outcome variables (see next question) **all participants in the policy treatment** group receive two questions to check whether they have properly read and understood the treatment.

Please answer the following two questions on the information you just received.

We gave you an example of an area where regulation aimed to limit harmful behavior: What was the example about?

[mining] [deforestation] [fishing] [industrial production]

In this example, you also read about a study that evaluated the policy's effectiveness. What did the study find?

The policy helped restore the health of the ecosystem, which had previously been threatened.

[correct] [not correct]

Detailed description of the outcome variable "real-world support for a price of carbon" and especially its real-world impact.

At the end of the policy description, respondents are asked a Yes/No question on whether they support the implementation of the described policy. This question measures respondents' **stated preferences** for the implementation of a social price of carbon via emission trading systems. All respondents have to answer this question.

All things considered, would you support the implementation of an emissions trading system for carbon as described above?

[Yes][No][don't know]

Afterwards, we measure **real-world support** for carbon pricing by asking respondents to choose the amount of money they would donate to the Citizens Climate Lobby, in case they were drawn to win USD 100.00 USD. Participants choose their donation using a slider that ranges from USD 0.00 to 100.00 and is pre-set at USD 50.00. All respondents have to answer this question.

By completing this survey, **you are automatically entered into a lottery to win USD 100.00**. In a few days, you will be notified if you have been selected. The payment will be made in the same way as your compensation for this survey, so no further action is required on your part.

If you are selected, you also have the option to donate a portion of your additional compensation to a charity that works to implement a carbon pricing system in the U.S.: the Citizens' Climate Lobby (CCL)*.
*The Citizens' Climate Lobby (CCL) advocates for climate policies, with a strong focus on carbon pricing, through grassroots lobbying and bipartisan engagement with lawmakers. Their efforts have contributed to successful state-level carbon pricing initiatives, such as California's cap-and-trade system and Oregon's 2023 carbon pricing law, while also pushing for similar action at the federal level. CCL's commitment to transparency is reflected in its Candid Platinum Seal of Transparency and the public

accessibility of its Form 990s.

Should you win the lottery, please enter your donation amount using the slider below:

[USD 0.00] ---------- 50 ----------[USD 100.00]

Please indicate in 3-5 sentences whether and why the conditions for the shortcut IRB apply to your experiment.

Our treatment does not involve any measures of physical states and does not pose physical risks to participants. Participation is fully voluntary and opportunity costs of time are compensated. Participants are properly informed about the duration of the experiment, the potential benefits of participation, and data security regulations.

Participants are not deceived. Detailed information about the sources for the information provided to respondents is be given at the end of the survey.

Please specify your incentive structure.

We use a lottery incentive in which one participant wins a prize of USD 100.00 to get a "real world outcome measure" of policy support. People can donate parts or all of their potential winnings.

Is the treatment effect identifiable using OLS, a probit/logit, or an ordered-logit model?

Yes.

For the continuous main outcome variable, the donation amount, we identify the treatment effect by including a binary indicator that indicates membership in the treatment group (treatment-dummy) in a linear regression model (OLS). This model can (but must not) include further individual our household-level predictors. The inclusion of further predictors may improve precision in the case of improper randomization.

The coefficient estimate of the treatment dummy measures the difference in the donation amount between treatment and control group.

For the binary outcome variable that measures stated preferences (Yes/No) we identify the treatment effect by including a binary indicator that indicates membership in the treatment group (treatment-dummy) in a linear probability model using OLS. The coefficient estimate of the dummy variable measures the difference in the likelihood of supporting the policy between the treatment and the control group.

What is the estimated length of the experiment? (excl. policy survey)

6.0 minutes.

Is your proposal using incentives?

Yes, as described above participants can choose to donate their additional incentives or keep them for themselves. The additional incentive amounts to USD 100.00 / 350 participants i.e. GBP 79.00 / 350 participants.

What is the average payment per participant (including flat fee + potential bonus payment)?

USD 2.00 (flat fee) + USD 0.29 (bonus) = USD 2.29 (total costs)

Which experimental software do you plan to use?

Qualtrics.

What is your expected effect size? (Cohen's d for continuous outcome variables and Cohen's h for shares/proportions)

Cohen's D: 0.35

Cohen's H: 0.2